

Day 15 — TNPSC Group 2A Study Material

Part A – General Studies: Geography of India: Natural Calamity, Disaster Management, Environmental Pollution, Climate Change & Green Energy

READING MATERIAL

Final sub-topic of Geography Unit II. Real exam evidence favors hazard-classification matching and statement-evaluation about disaster-management institutions (especially Tamil Nadu's own SDMA) over generic climate-change theory.

Q: Real exam: Match each hazard type to its example — Natural Hazard, Social Hazard, Biological Hazard, Technological Hazard.

A: Natural Hazard → Tsunami. Social Hazard → War. Biological Hazard → Bird Flu. Technological Hazard → Oil spill.

Q: Real exam: Evaluate these statements about Tamil Nadu's State Disaster Management Authority (SDMA) — (i) it coordinates disaster response and risk reduction (ii) the Chief Secretary chairs it (iii) the state cabinet follows NDMA guidelines when approving the state disaster management plan.

A: Only (i) is correct. (ii) is false — TN's SDMA is chaired by the Chief Minister, not the Chief Secretary. (iii) is also false as stated in that exact form.

Q: What do the acronyms SFDRR and NEOC stand for (the two CONFIRMED-correct pairings from a real, though TNPSC-flagged-as-disputed, exam question)?

A: SFDRR = Sendai Framework for Disaster Risk Reduction. NEOC = National Emergency Operation Committee. (The same question's other two pairings — SDMA as 'State Disaster Management Action' and SDMP as 'State Disaster Management Project' — were marked incorrect by the exam, but the question itself was flagged by TNPSC for expert-committee review, so treat the 'correct' full forms — likely 'Authority' for SDMA and 'Plan' for SDMP — as needing independent verification rather than settled fact.)

Q: What are the main types of environmental pollution (recap from the Unit I Environment/Ecology topic, now applied at the India-specific geography level)?

A: Air, water, soil, noise, and thermal pollution — each tied to specific Indian geographic contexts (e.g., industrial belts, river pollution from urban/industrial discharge, e-waste hotspots).

Q: Which renewable energy source has the largest installed grid-interactive capacity in India?

A: Wind power.

Q: What is ocean thermal energy, and how does it generate electricity?

A: It generates electricity by exploiting the temperature DIFFERENCE between warmer surface ocean water and colder deep water — the temperature gradient drives the power generation process, rather than direct solar or wind capture.

MOCK TEST (WITH ANSWERS)

1. Which type of hazard is a Tsunami classified as?

- A. Social Hazard
- B. Biological Hazard
- C. Technological Hazard
- D. Natural Hazard ✓**

Explanation: Natural Hazard. [easy, question-bank-2026]

2. Which type of hazard is an Oil Spill classified as?

- A. Natural Hazard
- B. Social Hazard
- C. Biological Hazard
- D. Technological Hazard ✓**

Explanation: Technological Hazard. [easy, question-bank-2026]

3. In Tamil Nadu's State Disaster Management Authority (SDMA), who acts as the chairperson?

- A. The Chief Secretary
- B. The Chief Minister ✓**
- C. The Governor

D. The District Collector

Explanation: The Chief Minister chairs it — a real exam marked the 'Chief Secretary' claim as false. [hard, question-bank-2025]

4. SFDRR stands for:

A. State Framework for Disaster Risk Reduction

B. Sendai Framework for Disaster Risk Reduction ✓

C. State Fund for Disaster Relief and Rehabilitation

D. Sendai Fund for Disaster Risk Reduction

Explanation: Sendai Framework for Disaster Risk Reduction. [medium, question-bank-2025]

5. Which renewable energy source has the largest installed grid-interactive capacity in India?

A. Solar power

B. Wind power ✓

C. Small hydro

D. Biomass

Explanation: Wind power. [medium, web-research]

6. Ocean thermal energy generates electricity using:

A. Wave height differences

B. Tidal cycles

C. Temperature differences at different ocean depths ✓

D. Underwater wind currents

Explanation: Temperature gradient between surface and deep water. [medium, web-research]

Part B – Aptitude & Mental Ability: Alphanumeric Reasoning

READING MATERIAL

Alphanumeric Reasoning is 1 of 6 named Reasoning skills, combining letters and numbers in a single pattern. Real exam evidence (independently re-verified) shows three core techniques: letter-to-number position coding, variable-gap letter sequences, and fixed-shift word ciphers.

Q: Method — letter-to-number position coding: If ACEG is coded as 16, and DFGH is coded as 25, how would HIKM be coded?

A: Each letter is converted to its position in the alphabet (A=1, B=2, ... Z=26) and summed. ACEG: A=1, C=3, E=5, G=7, sum=16 ✓. DFGH: D=4, F=6, G=7, H=8, sum=25 ✓. Applying the same rule to HIKM: H=8, I=9, K=11, M=13, sum=8+9+11+13=41.

Q: Method — widening-gap letter sequence: Find the next letter: B, E, I, N, ?

A: Check the gaps between consecutive letters' positions: B(2)→E(5) is +3, E(5)→I(9) is +4, I(9)→N(14) is +5. The gap itself increases by 1 each time, so the next gap is +6: N(14)+6=20, which is T.

Q: Method — fixed-shift word coding (Caesar-cipher style): If ROAD is coded as URDG, how would SWAN be coded using the same rule?

A: Check the shift for each letter: R→U is +3, O→R is +3, A→D is +3, D→G is +3 — a uniform +3 shift applied to every letter. Applying the same +3 shift to SWAN: S→V, W→Z, A→D, N→Q, giving VZDQ.

Q: Method — two-interleaved-sequences letter pair: A pattern alternates first-letters increasing by a fixed gap and second-letters decreasing by the same gap (e.g., AZ, GT, MN, ?, YB). How do you solve it?

A: Track the two positions (first letter, second letter) as SEPARATE sequences. First letters: A(1), G(7), M(13) — increasing by 6 each time, so next is 19=S, then 25=Y (matches the given final term Y). Second letters: Z(26), T(20), N(14) — decreasing by 6 each time, so next is 8=H, then 2=B (matches the given final term B). So the missing term is S+H = 'SH'.

MOCK TEST (WITH ANSWERS)

1. If BDFH is coded as 20 (sum of letter positions), what would CEGI be coded as using the same rule?

- A. 22
- B. 24 ✓**
- C. 26
- D. 28

Explanation: C=3, E=5, G=7, I=9, sum=24. (Check rule: B=2, D=4, F=6, H=8, sum=20 ✓ matches given.) [medium, authored]

2. Find the next letter in the sequence: C, F, J, O, ?

- A. S
- B. T
- C. U ✓**
- D. V

Explanation: Gaps: C(3)→F(6) is +3, F(6)→J(10) is +4, J(10)→O(15) is +5, so next gap is +6: O(15)+6=21=U. [medium, authored]

3. If PLAN is coded as SODQ using a fixed letter shift, how would TRIM be coded using the same rule?

- A. WULP ✓**
- B. WUKP
- C. VULP
- D. WULO

Explanation: Shift check: P→S, L→O, A→D, N→Q — a uniform +3 shift. Applying +3 to TRIM: T→W, R→U, I→L, M→P, giving WULP. [hard, authored]

4. In the sequence DY, IT, NO, ?, XE, the first letters increase and second letters decrease by a fixed gap. What is the missing term?

- A. SJ ✓**
- B. SI
- C. TJ
- D. RJ

Explanation: First letters: D(4), I(9), N(14) — gap +5, so next is 19=S, then 24=X (matches given X). Second letters: Y(25), T(20), O(15) — gap -5, so next is 10=J, then 5=E (matches given E). Missing term: SJ. [hard, authored]

Part C – General English: Poem: "On Killing a Tree" — Gieve Patel

READING MATERIAL

A poem arguing that truly killing a tree takes far more than a single cut — it requires destroying its root, the hidden source of its life. Real coaching-site evidence (after discarding one mismatched/wrong-poem question found on the source page) tests figure-of-speech identification on specific lines.

Q: What is the poem's central argument?

A: It takes much more effort to truly kill a tree than a single jab or cut. The tree has grown slowly, absorbing years of sunlight, air, and water — cutting and bruising it isn't enough, since it will heal and grow again. The root, hidden underground, must be exposed and fully destroyed for the tree to actually die.

Q: What is the deeper/metaphorical meaning some readings draw from the poem?

A: The poem can be read as being about confronting deep-seated feelings or habits — superficial treatment (a quick cut) doesn't resolve something with deep roots; only fully exposing and addressing the root cause truly 'kills' it.

Q: Identify the figure of speech: "It has grown / Slowly consuming the earth."

A: Personification — the tree is given the active, almost intentional action of 'consuming' the earth, as though it were a living agent in pursuit of nourishment.

Q: Identify the figure of speech: "leperous hide" (describing the tree's bark).

A: Metaphor — the bark is directly described as 'leperous hide' (skin), without 'like/as', comparing its rough, scarred texture to a disease-marked skin.

Q: Identify the figure of speech: "The bleeding bark will heal."

A: Personification — the bark is described as capable of 'bleeding' and 'healing', human/animal-like properties applied to plant tissue.

Q: Identify the figure of speech: "It is to be roped, tied, / And pulled out — snapped out / Or pulled out entirely."

A: Repetition — 'pulled out' is repeated, emphasizing the forceful, thorough effort needed to fully remove the root.

Q: Identify the figure of speech: "Out of the anchoring earth."

A: Metaphor — the earth is described as 'anchoring', directly likening it to an anchor holding the root in place, without 'like/as'.

MOCK TEST (WITH ANSWERS)

1. According to the poem, what is required to TRULY kill a tree (not just cut it)?

- A. Burning its leaves
- B. A single deep cut
- C. Exposing and destroying its root ✓**
- D. Removing its fruit

Explanation: The root, hidden underground, must be exposed and destroyed — a cut alone allows the tree to heal and regrow. [medium, web-research]

2. "It has grown / Slowly consuming the earth" uses which figure of speech?

- A. Simile
- B. Metaphor
- C. Personification ✓**
- D. Hyperbole

Explanation: Personification — the tree given the active action of 'consuming'. [medium, web-research]

3. "Leperous hide" describing the tree's bark is an example of:

- A. Simile
- B. Metaphor ✓**
- C. Personification
- D. Alliteration

Explanation: Metaphor — bark directly described as diseased skin. [medium, web-research]

4. "The bleeding bark will heal" uses which figure of speech?

- A. Metaphor

B. Hyperbole

C. Personification ✓

D. Simile

Explanation: Personification — bark given the ability to 'bleed' and 'heal'. [medium, web-research]

5. The repeated phrase "pulled out" in the lines about the root is an example of:

A. Anaphora

B. Epistrophe

C. Repetition ✓

D. Alliteration

Explanation: Repetition — emphasizing the forceful effort needed. [medium, web-research]

6. "Out of the anchoring earth" uses which figure of speech?

A. Simile

B. Metaphor ✓

C. Personification

D. Alliteration

Explanation: Metaphor — earth described directly as 'anchoring'. [medium, web-research]